Olist Commerce360

An End-to-End E-Commerce Data Warehouse and Business Intelligence Platform

P1_01 - Project Proposal

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Course / section	DL3004 - Data Warehousing and Business Intelligence - [Section]
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1. Background

Olist is a Brazilian marketplace integrator. Small and medium sellers list their products through Olist, which sells them on large Brazilian marketplaces and arranges delivery through logistics partners. Olist published an anonymised extract of its operational data on Kaggle (*Brazilian E-Commerce Public Dataset by Olist*). It contains 99,441 orders placed between 4 September 2016 and 17 October 2018, together with their items, payments, customer reviews, products, sellers and customer locations.

2. Business problem

The data is stored as nine separate operational tables, all linked through *order_id*. Sales value is recorded per item, money received is recorded per payment, delivery milestones per order, and satisfaction per review. Each of these has a different number of rows per order. Management cannot reliably answer cross-cutting questions such as *which regions receive late deliveries*, *which categories carry the highest freight burden* or *whether late orders receive lower review scores*, because a direct join of the tables repeats values. In the source data, joining items to payments raises the total amount paid from BRL 16.01 million to BRL 20.31 million.

The project will build a dimensional data warehouse that integrates these processes at their correct grain. The warehouse will support a Power BI dashboard for decision-making.

3. Objectives

- **O1 (Phase 1):** Analyse business requirements and design a dimensional model. The model will have documented grain, keys, hierarchies and measures for four business processes: sales, order fulfilment, payments and reviews.
- **O2 (Phase 2):** Implement the model in PostgreSQL with *project_staging*, *project_dw* and *project_reporting* schemas. Load it with reproducible SQL/Python ETL, surrogate keys, and data-quality checks that reconcile row counts and value totals with the source.
- **O3 (Phase 3):** Provide analytical SQL, ROLLUP/CUBE OLAP queries and reporting views. Optimise queries with indexes (EXPLAIN ANALYZE before and after), and build a three-page Power BI dashboard with evidence-based recommendations.

4. Scope

In scope: the four business processes below, analysed by date, customer, product, seller, payment type and order status.

Business process	Fact table	Fact type
Sales (order line items)	fact_order_items	Transaction
Order fulfilment lifecycle	fact_orders	Accumulating snapshot
Payment transactions	fact_payments	Transaction
Customer reviews	fact_reviews	Transaction

Out of scope (the source does not contain the required data):

Analysis	Reason	Feasible alternative
Profit / margin by product or seller	No cost of goods, commission or carrier cost columns.	Revenue, freight-to-price ratio, AOV.
Inventory movement / stock snapshots	No stock level or warehouse data.	Units sold and demand trend by category.
Promotion / campaign effectiveness	No promotion, coupon campaign or marketing channel data.	Voucher usage share in payments (descriptive only).
Customer demographics (age, gender, income)	Not in source.	Behavioural segmentation: region, order frequency, spend.
Product-level or seller-level review analysis	Reviews are given per order, not per item; 1.3% of orders have several sellers.	Order-level review analysis; seller view limited to single-seller orders if required later.
Order status history	Only the final status is provided.	Milestone timestamps in fact_orders.

5. Data source

Kaggle: *olistbr/brazilian-ecommerce*, version 2 (https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce). The licence is shown on Kaggle as CC BY-NC-SA 4.0. The data is used for non-commercial academic work with attribution, and the raw files are not redistributed in the project repository. Row counts below were verified with the profiling script *02_Data/profiling/profile_source.py*.

File	Rows	Content
olist_orders_dataset.csv	99,441	One row per order: customer, status, 5 lifecycle timestamps
olist_order_items_dataset.csv	112,650	One row per item line: product, seller, price, freight
olist_order_payments_dataset.csv	103,886	One row per payment record: type, instalments, value
olist_order_reviews_dataset.csv	99,224	One row per review: score, comment, dates
olist_customers_dataset.csv	99,441	One row per customer_id (per order); 96,096 distinct persons
olist_products_dataset.csv	32,951	Product category and physical attributes
olist_sellers_dataset.csv	3,095	Seller location
product_category_name_translation.csv	71	Portuguese to English category names
olist_geolocation_dataset.csv	1,000,163	Latitude/longitude points per zip prefix (19,015 prefixes)

The expected warehouse volumes are about 112,650 fact rows in fact_order_items and 99,441 in fact_orders. dim_customer will hold about 96,097 rows and dim_product about 32,952. These exceed the course minimum of 100+ dimension records and 10,000+ fact records. Actual loaded counts will be verified in Phase 2.

6. Proposed architecture

Layer	Description
1. Source	9 Kaggle CSV files in 02_Data/raw_data (not committed to Git).
2. Staging - project_staging	One stg_ table per CSV loaded as text; type conversion, trimming, zip padding, de-duplication.
3. Cleaning & transformation	Python/pandas and SQL: unknown members, lookups, derived attributes, per-order aggregation.
4. Warehouse - project_dw	6 dimensions with surrogate keys and 4 fact tables (this design).
5. Reporting - project_reporting	Views for analysis (e.g. vw_monthly_sales) and OLAP queries (ROLLUP/CUBE).
6. BI - Power BI Desktop	Fact and dimension tables imported separately with 1:* relationships; 3 dashboard pages.

7. Stakeholders

Executive management, category managers, marketplace/seller relations, logistics, customer experience, finance, marketing/CRM and the BI team. Their needs are detailed in the Business Requirements Document (P1_02).

8. Deliverables and timeline

Phase (weight)	Weeks	Main deliverables
1 - Business analysis & dimensional design (30%)	1-3	Proposal, BRD, business questions, fact/dimension identification, grain definition, star schema diagram, data dictionary
2 - Physical DW & ETL (40%)	4-6	PostgreSQL schemas and DDL, staging, dimension and fact load scripts, ETL scripts, data cleaning report, data quality report
3 - OLAP, optimisation & BI (30%)	7-8	Analytical and OLAP SQL, reporting views, indexing report, Power BI dashboard, screenshots, insights report, presentation

9. Team roles

Role	Main responsibility (all members review and must be able to explain the full model)
Member 1 - Warehouse & integration lead	PostgreSQL implementation, load scripts, surrogate keys, integrity checks, indexing
Member 2 - Business analysis & modeling lead	Requirements, dimensional model, cleaning rules, business insights
Member 3 - Analytics & BI lead	Business questions, data-quality report, OLAP/reporting SQL, Power BI dashboard

10. Tools

PostgreSQL and pgAdmin, Python 3 with pandas, SQL, Power BI Desktop, dbdiagram.io / draw.io, and GitHub for version control. No paid infrastructure is required.

11. Main risks and mitigation

Risk	Mitigation
Double counting when orders, items and payments are joined	Separate facts at their own grain; per-order aggregates are computed before joining (P1_05).
customer_id changes with every order	dim_customer is keyed on customer_unique_id (96,096 persons).
Missing or out-of-sequence timestamps	Unknown date key -1; lags left NULL; sequence errors are flagged, not deleted.
Very few orders in Sep-Dec 2016 and Sep-Oct 2018	Rows are kept; trend and growth analysis uses Jan 2017 - Aug 2018.
Inconsistent documents across team members	One model specification generates the diagram, dictionary and tables; decisions are logged.